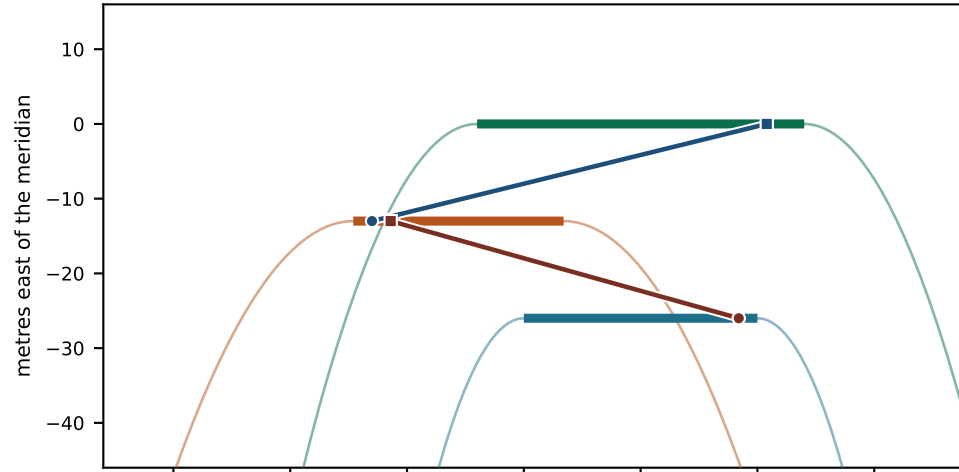


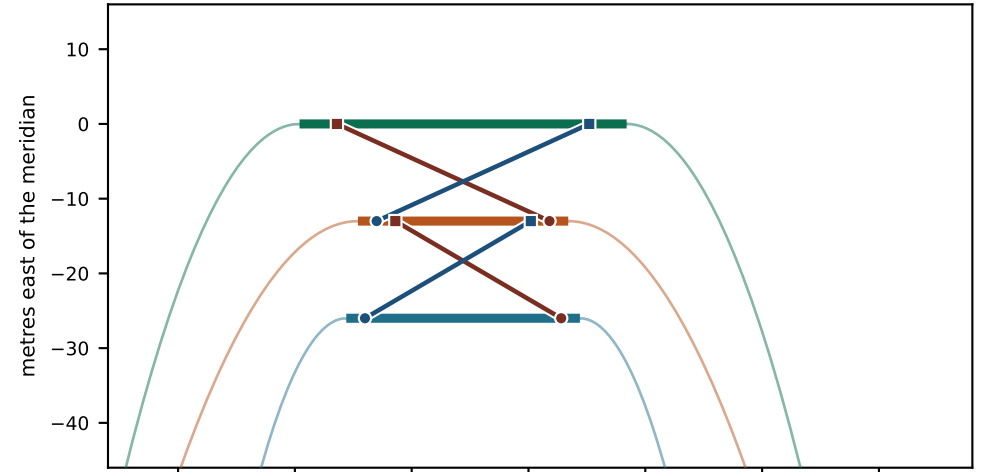
# Transfer lines between the rings

the east straights are the only place two rings run parallel and close, so every transfer is a translation: two opposite bends,  $2d/L$  of bending for an offset  $d$  over a line of length  $L$

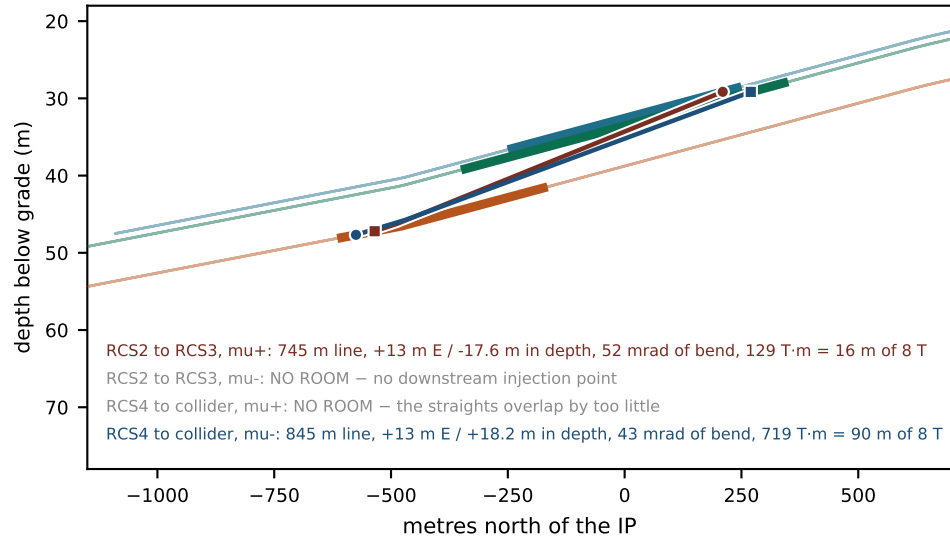
(a) as built: RCS3/4's straight sits 390 m south of the other two



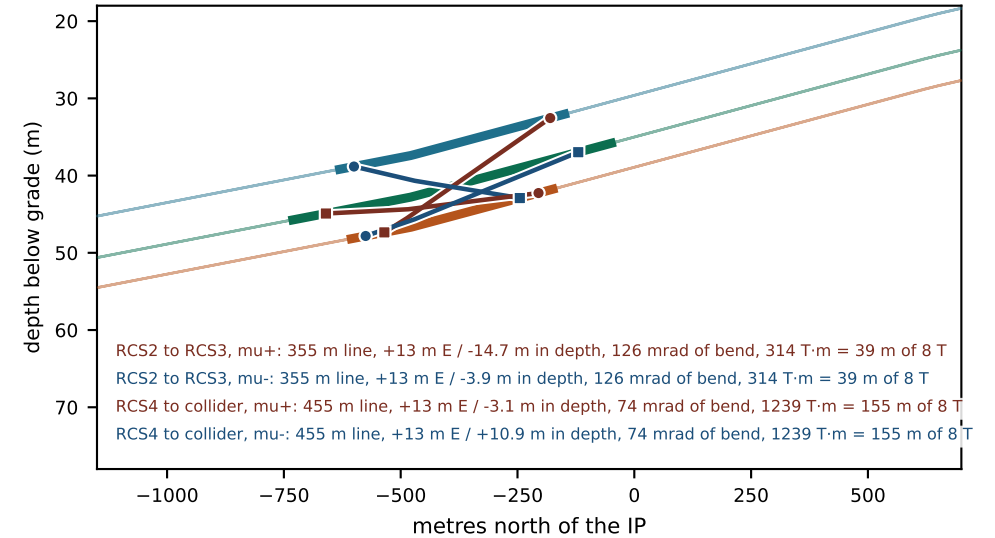
(b) concentric: the collider and RCS1/2 slid 390 m south onto RCS3/4's centre



(c) the same region in section



(d) the same region in section



$\mu^+$  transfer line (south-going on the east straights)  
 $\mu^-$  transfer line (north-going)

● extraction septum  
■ injection septum  
— east straight  
— the arcs peeling off it

RCS1 → RCS2 and RCS3 → RCS4 stay inside their own tunnel (vertically stacked rings, a 1.5 m dogleg over 200 m: 16 and 75 T·m, 2 and 9 m of 8 T) and are not drawn.

$\mu^+$  and  $\mu^-$  counter-rotate, so each transfer needs a mirror pair of lines. As built they are not mirrors: RCS3/4's straight is 390 m south of the other two, so for one sign of each transfer the two straights barely overlap and there is no room for a line. Sliding the collider and RCS1/2 onto RCS3/4's centre restores the symmetry – all four lines close, at roughly twice the bending each, and the collider's north-arc cover improves from 5.9 m to 12.9 m.